

Ultrasound Reporting System

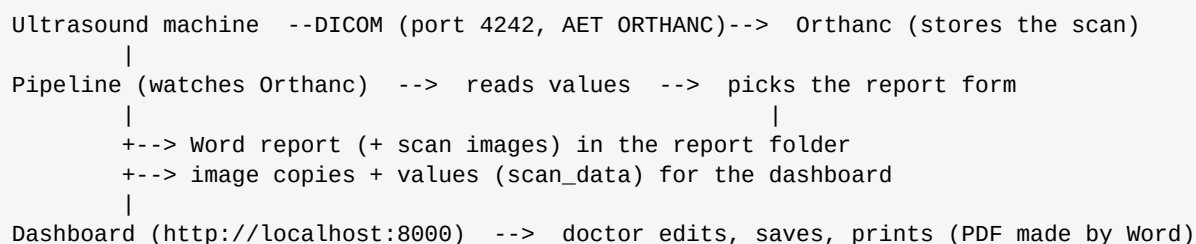
Technical Guide - installation, updates and maintenance

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For the person who installs and maintains the system. Clinic staff use the separate *Staff and Doctor Guide*.

1. How the system works

Everything runs on one Windows PC in the clinic, next to the ultrasound machine.



- 1 The scanner sends a study to **Orthanc** (a free DICOM server, Windows service).
- 2 When no image has arrived for StableAge seconds (15), Orthanc marks the study *stable*. The **pipeline** polls Orthanc's change log every 3 s and picks it up.
- 3 It reads the measurements: the scanner's **Structured Report** (labelled values: BPD, EFW, Uterus Length...) and, with **Tesseract OCR**, the values printed on each image.
- 4 It decides the report type (exam/preset chosen on the scanner, gestational age, measurements) and fills that type's **Word form**. The report, with the first 6 images, is saved in the report folder.
- 5 The **dashboard** lists the scans by day. The doctor edits the Word report in the browser; it is saved back into the same .docx. **PDF / Print** has Microsoft Word convert the .docx to PDF, so print and Word match.

2. Components

File / folder	Purpose
pipeline.py	Watches Orthanc, reads each study, builds the report data, records it for the dashboard. Also manual commands: run.bat list / study <id> / folder <dir> / search
orthanc_client.py	Orthanc REST client: studies, instances, change log, delete
extract_measurements.py	Measurements from DICOM Structured Reports
ocr_extract.py	Values printed on the images (Tesseract)
scan_classifier.py	Which report type a study is
fill_report.py	Fills a Word form ({{{placeholders}}}), adds the images page, saves the report
measurement_rules.py	Suggests the organ for unlabelled distance values (abdomen/pelvis)
report_generator.py	Measurements-only PDF, for a study whose type is not recognised
scan_store.py	Where each scan's image copies and values (scan_data) are kept
docx_pdf.py	Word report to PDF with Microsoft Word (PowerShell COM)
webapp/	Dashboard (FastAPI): worklist, editor (docx_editor.py), images (report_images.py), settings, database (db.py)
templates/	Default Word forms, one per report type (see section 8)
tools/build_templates.py	Builds the default Word forms
deploy/	Setup, release, service and maintenance scripts (sections 5-7)
setup.bat	Installer / updater (section 6)
Restart Services.bat	Restarts Orthanc, pipeline and dashboard and checks them

3. Folders and files

Installed copy: C:\scanning-machine. setup.bat replaces the program files on an update, but never the clinic's own files (marked **kept**).

Path	Contents	On update
config_local.py	This clinic's settings (section 4)	kept
data\app.db	Dashboard database: scans, reports, status, Settings page values	kept
data\forms\	This clinic's Word forms (copied from templates\ when first needed)	kept
data\session_secret.txt	Dashboard login cookie key	kept
watch_state.json	Last Orthanc change processed - without it old studies are processed again	kept
reports\filled\	Reports when no report folder is set (or it is unreachable)	kept
reports\scan_data\	Image copies when no report folder is set	kept
logs\	pipeline.log, dashboard.log (rotated at 10 MB)	kept
.venv\	Python packages for this install	updated
Everything else	Program files	replaced

Report folder (dashboard **Settings**), for example D:\GeneratedReports\reports: the Word reports and their PDFs, and scan_data\<orthanc id>\ with every image of each scan (JPEG) and scan.json (values per image, scanner report values).

Orthanc keeps the original DICOM in its own StorageDirectory (here C:\Orthanc), configured in C:\Program Files\Orthanc Server\Configuration\orthanc.json.

4. Settings

config_local.py (Notepad; restart the services after a change). It only needs the values that differ from config.py. setup.bat writes the first ones.

Setting	Meaning	Typical
ORTHANC_URL	Orthanc's web address. Use 127.0.0.1, not localhost (IPv6 delay)	http://127.0.0.1:8042
ORTHANC_USERNAME / _PASSWORD	Orthanc login, when its authentication is on	
POLL_INTERVAL_SECONDS	How often the pipeline checks Orthanc	3
TESSERACT_CMD	Path of tesseract.exe	C:\Program Files\Tesseract-OCR\tesseract.exe
DASHBOARD_LOGIN_REQUIRED	False: no login, dashboard answers only on this PC. True: login, open to the LAN	False
ORTHANC_KEEP_DAYS	Days before a reviewed scan is deleted from Orthanc; 0 = never (section 11)	30
MAX_REPORT_IMAGES	Images put in a new report	6
HEARTBEAT_URL	healthchecks.io ping URL - alerts when the PC stops working	

Setting	Meaning	Typical
CLINIC_NAME / _ADDRESS / _PHONE	Printed on the measurements-only PDF	

Dashboard Settings page (stored in the database): reporting doctor name and qualification (PCPNDT declaration and signature - a report is flagged while it is empty), default *Ref by*, and the **report folder**. The folder is checked for writing before it is accepted; saving it also moves image copies still on C: into it.

5. Services and accounts

What	How it runs	Account
Orthanc	Windows service Orthanc , automatic start	Local System
Pipeline	Scheduled task Ultrasound Pipeline : at boot (+1 min), runs <code>deploy\service_pipeline.bat</code> , which restarts <code>pipeline.py</code> watch 30 s after any exit	SYSTEM
Dashboard	Scheduled task Ultrasound Dashboard : at boot, runs <code>deploy\service_dashboard.bat</code> (uvicorn, port 8000), same restart loop	the Windows user, S4U (no stored password)

Important: the dashboard must run as a real Windows user: Windows refuses to start Microsoft Word for SYSTEM (COM error 0x80070005 *Access is denied*), so PDF / Print fails. `setup.bat` sets this up; to redo it, run `deploy\dashboard_as_user.bat` and press Enter (S4U: runs without anyone signed in, needs no password, survives password changes). If S4U ever fails, type the user's password instead - then the script must be run again after every password change. A user with a blank password cannot use the stored-password option (policy `LimitBlankPasswordUse`).

Restart Services.bat (Desktop: *Restart Ultrasound Services*) asks for admin rights, stops both tasks and any leftover python processes, restarts Orthanc, starts the tasks, waits for Orthanc (:8042), the dashboard (:8000) and the pipeline log line *Watching Orthanc*, then shows OK / FAILED for each.

6. Installing a new clinic

Needs: Windows 10/11 64-bit, internet during setup, administrator rights, Microsoft Word installed and activated.

Before setup

- Orthanc:** install from `orthanc-server.com` (Windows installer). In `orthanc.json` set "StableAge" : 15 (default 60 s delays every report); keep "DicomAet" : "ORTHANC", "DicomPort" : 4242. If remote access is allowed, change the default orthanc/orthanc password. Then `Restart-Service Orthanc`.
- Release zip:** on the development PC, commit, then run `deploy\make_release.bat`. It packs the last commit (git archive, no local data) into `dist\scanning-machine-<commit>.zip`.

setup.bat

Extract the zip anywhere, double-click **setup.bat**, click Yes. It:

- stops a running pipeline/dashboard (on updates);
- copies the program to `C:\scanning-machine` (never the files marked *kept* in section 3);
- installs Python 3.11 if no 3.11/3.12 is found, creates `.venv`, installs `requirements.txt`;
- asks: Orthanc URL / username / password (tested at once), clinic name, address, phone, heartbeat URL, and whether the dashboard is **used only on this PC** (yes = no login, no firewall opening);

- opens the firewall for DICOM 4242 (and 8000 when the dashboard is used from other PCs) to the LAN and Tailscale only;
- disables sleep on mains power; installs Tesseract if missing; checks Word;
- registers and starts both scheduled tasks, then runs dashboard_as_user (press Enter);
- puts *Ultrasound Dashboard* and *Restart Ultrasound Services* on the shared Desktop;
- prints the dashboard address (and the first-login setup token when a login is used).

After setup

- 1 Dashboard **Settings**: reporting doctor, qualification, default Ref by, report folder (e.g. on D:).
- 2 On the ultrasound machine: DICOM destination = this PC's LAN IP, port **4242**, AE title **ORTHANC**; press Verify. Give this PC a fixed IP (or DHCP reservation) so the destination stays valid.
- 3 Send a test study: the report appears in about 30 s. Open it, add an image, save, **PDF / Print**.
- 4 Restart the PC once and check everything comes back by itself.

7. Updating and development

Development happens in a **separate git folder** (on this PC `C:\Donttouch\scanning-machine`); the clinic runs the installed copy. Changes in the development folder are **not live** until installed.

- 1 Change and test in the development folder (a second dashboard can run on another port: `.venv\Scripts\python -m uvicorn webapp.main:app --port 8001`).
- 2 Commit and push. Optionally run `tools/build_templates.py <type>` first if a form changed.
- 3 **This PC**: double-click `deploy\move_to_install.bat` in the development folder. It builds the release from the last commit and runs its `setup.bat`; the existing data is kept (answer **Y** to keep the configuration).
- 4 **Another clinic**: `deploy\make_release.bat`, copy the zip over (e.g. AnyDesk), extract, run `setup.bat`, answer **Y**.
- 5 Check the version setup prints (VERSION = commit and date), then send a test study.

Tip: update when someone at the clinic can confirm it works, not before a weekend. To roll back, run `setup.bat` from the previous zip - keep the last one or two.

8. Report forms

Type key	Form	Chosen when
early_pregnancy	Early pregnancy	OB exam, GA under 11 weeks
nt_scan	NT scan	OB exam, GA 11-14 weeks
anomaly_scan	Anomaly scan	OB exam, GA 14-28 weeks
growth_scan	Growth scan	OB exam, GA 28 weeks and over
follicular_study	Follicular study	preset GYN_FERTILITY
abdomen_pelvis_female / _male	Abdomen & pelvis	abdomen exam or GYN preset; by patient sex
breast_scan	Breast	exam or preset containing BREAST

Each form is a normal Word file with `{{placeholders}}` where values go (`{{patient_name}}`, `{{bpd}}`, `{{uterus_size}}`...). A placeholder with no value prints as `____` so the doctor cannot miss it (patient-box fields stay empty).

- **Defaults** live in `templates\` (in git). `python tools/build_templates.py breast_scan` rebuilds one form; with no name it rebuilds **all** and overwrites hand edits - avoid.

- **Each clinic** uses data\forms\, copied from the default the first time a type is used. Edit those in Word; updates never replace them. To give a clinic an improved default, copy it into data\forms by hand (or delete their copy to get the default).
- **New report type:** add a builder in tools/build_templates.py (FORMS), the key to SCAN_TYPES in fill_report.py and SCAN_TYPE_NAMES in webapp/main.py, and a rule in scan_classifier._classify_from_exam.

9. How values reach the report

- **Labelled values** (Structured Report): mapped by name to form fields in pipeline._MEASUREMENT_FIELD_MAP, converted to the unit printed on the form (_FIELD_UNITS). A value whose unit cannot be converted is left blank and the report flagged - a blank is safe, a wrong number is not.
- **Organ sizes:** Length / Width / Height of uterus, ovaries, kidneys, prostate are combined into one "L x W x H cm". Prostate volume = L x W x H / 2 when not measured.
- **Derived:** EFW rounded to grams with $\pm 14.6\%$ (error of the scanner's Hadlock 1985 formula, as in the clinic's reports); FL/AC % from FL and AC; gestational ages split into weeks/days; EDD/LMP dates when sent.
- **Unlabelled distances** (plain calliper, "Dist 9.70 cm"): not filled automatically. measurement_rules.py suggests the organ from the order and sizes; the doctor confirms with a click.
- **Values printed on images** (OCR): shown under each image in the dashboard to click into the report.
- **Name and age:** an age typed after the name on the scanner ("LAKSHMI 27") is split off into Age.

10. Review flags

A report that was made but looks questionable is marked *Needs checking* with the reason:

Reason	Cause
scan type not identified / low classification confidence	No clear exam/preset; check the form, use Remake report
no measurements extracted / no images captured	Nothing labelled was sent, or only SR
N measurement(s) left blank - units could not be reconciled	Unit conversion failed (section 9)
doctor name not set in Settings	PCPNDT declaration would print blank
scan dated X but received Y	Study date more than 1 day from arrival: scanner clock wrong (or an old study re-sent)
report folder ... not reachable - saved on this PC	Report folder could not be written; saved in reports\filled

11. Storage and clean-up

Where	Per scan (approx.)
Orthanc DICOM (C:\Orthanc)	12 MB
Word report + PDF (images inside)	3 MB
Image copies for the dashboard (scan_data)	1 MB

Once a day the pipeline deletes from Orthanc each study whose report the doctor has **reviewed** and that arrived more than ORTHANC_KEEP_DAYS (30) days ago - only if its image copies exist. Reports and image copies are kept; only *Remake report* stops working for that scan. Unreviewed scans and studies

unknown to the dashboard are never deleted. Keep the report folder on a large drive and back it up together with data\.

Check: PCPNDT requires obstetric records to be kept for two years. Confirm with the doctor that the reports and image copies are sufficient, or raise ORTHANC_KEEP_DAYS.

12. Troubleshooting

Start with the Desktop **Restart Ultrasound Services**; it says which part failed. Then read logs\pipeline.log / logs\dashboard.log.

Symptom	Check / fix
No report after a scan	Study in Orthanc? (http://localhost:8042 , or run .bat list). If not: scanner cabled and on the LAN, destination IP/port/AET, Verify on the scanner. If yes: pipeline.log.
Scanner Verify partly fails	Normal for the Philips 3D presentation object; 2D images still arrive.
PDF / Print: <i>Could not create the PDF</i>	dashboard.log: <i>Access is denied</i> on Word.Application = dashboard runs as SYSTEM: run deploy\dashboard_as_user.bat. Otherwise Word not installed/activated, or a Word dialog.
Dashboard FAILED, port 8000 in use	Another process holds port 8000; the restart script stops it - restart again.
Pipeline log: <i>Waiting for Orthanc</i>	Orthanc service stopped, or wrong ORTHANC_URL / password.
Reports on the wrong day	Scanner date/time wrong (flagged "scan dated ... but received ...").
All old studies processed again	watch_state.json lost - restore it, or set its last_seq to the current Orthanc change number.
Images missing in the dashboard	Report folder unreachable or changed: check Settings; image copies may still be in reports\scan_data.
Remake report: images removed	Study already cleaned from Orthanc (section 11); edit the report instead.

13. Security

- Patient reports are medical records (PCPNDT). The dashboard is plain HTTP: never expose port 8000 or 8042 to the internet; remote support only through AnyDesk / Tailscale.
- With DASHBOARD_LOGIN_REQUIRED = False the dashboard refuses every connection not from the PC itself.
- Give the Windows user a password (with automatic sign-in if needed); a PC without one is open to anyone at it.
- Orthanc: keep remote access off, or change its default password.
- config_local.py holds the Orthanc credentials and is never put in git or in the release zip.

14. This clinic / open items

Clinic PC	Ganesh Healthcare, Kolar - DESKTOP-918QFN1, user ADMIN, 192.168.1.15
Scanner	Philips Affiniti 50, 192.168.1.100 (port 104), cabled; exports to 192.168.1.15 / 4242 / ORTHANC
Install	C:\scanning-machine; development folder C:\Donttouch\scanning-machine
Report folder	D:\GeneratedReports\reports (images in its scan_data)
Tasks	Pipeline as SYSTEM; Dashboard as ADMIN (S4U); dashboard used on this PC only (no login)
Old task backups	data\Ultrasound Pipeline.xml, data\Ultrasound Dashboard.xml (pre-install setup)

Open items

- Follicle measurements: the scanner's follicle labels have not been seen yet; the follicle table is filled by hand.
- Breast preset name not confirmed on the scanner (rule matches any preset containing BREAST).
- Doppler images carry PI/RI but no vessel name; the doctor clicks them into the right row.
- Possible later step: a single Setup.exe (Inno Setup) bundling Python, packages, Tesseract and Orthanc for offline installs.